Erasure-Boundary Expansions in a Fixed-Shift Native Gram: Mixed Words, Residual Parity, and Cross-Key Holonomy

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Abstract

Let F be the atomic-normalized missing-native incidence at one fixed nonzero shift and write its Gram as

$$F^*F = I + R + E.$$

Here R contains collisions already present at a common residual output (i, s), whereas E contains the unequal-residual cross terms created when s is erased inside a complete native key i. This paper resolves the exact algebra of mixed R/E operator words without asserting an arithmetic cancellation estimate.

First, we decompose $R = \sum_i R_i$ and $E = \sum_i E_i$ over complete native keys. The matrix R_i is block diagonal in the residual partition at i, while E_i is strictly off block. Consequently

$$\operatorname{tr}(E_i R_i^a) = 0 \qquad (a \geq 0).$$

In particular, the half-sector $\operatorname{tr}(ER)$ in the full mixed second-moment coefficient $S_{2,1} = 2\operatorname{tr}(ER)$ is not a one-key phenomenon:

$$\operatorname{tr}(ER) = \sum_{i \neq j} \operatorname{tr}(E_i R_j).$$

Its literal terms are two-key, two-row rectangles. After the source-row gauge is removed, their four target Möbius signs reduce to the two residual signs on the unequal-residual edge. The remaining complex coefficient is a gauge-invariant cross-key holonomy.

Second, we give the exact cyclic sector formula for the coefficient $S_{k,b}$ of t^b in $\operatorname{tr}(R + tE)^k$. The one-erasure sector is

$$S_{k,1} = k \operatorname{tr}(R^{k-1}E).$$

On every same-residual edge the residual sign factor is one; after the complete decorated walk is closed, all source-row gauges telescope. The remaining target-sign product is exactly the product of $\mu(s)$ over the odd-degree boundary of the multigraph of erasure-edge residual labels. Thus the number of erasure edges alone does not decide whether the signs freeze.

Third, although individual two-erasure words may be signed, their complete even-moment sector is nonnegative:

$$S_{2q,2} \geq 0.$$

We prove this by diagonalizing R and identifying the coefficient with the nonnegative divided difference of the odd increasing function x^{2q-1}. Hence the quadratic erasure response cannot supply a negative moment correction; any favorable signed response must enter through the linear sector or higher sectors. Local star compensation, cross-key countermodels, determinant reindexing, and the remaining arithmetic and shorted-reassembly gates are recorded explicitly. The matrix and sector identities are L0, their complete fixed-shift carrier and target-sign identifications are L1, and every uniform Möbius saving remains an unproved L2 input.

Keywords: fixed shift; native Gram; residual erasure; noncommutative word; Möbius sign; cross-key rectangle; divided difference; operator moment.

Convention. The arithmetic specialization uses one fixed nonzero shift h_0, the complete literal missing-high carrier, complete native output keys, and raw atomic normalization; the shift is not 2.

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1 Introduction

TPC–61 identifies the exact atomic-normalized native Gram of the missing-high carrier and splits its off-diagonal alias operator into same-residual and unequal-residual parts [3]. TPC–68 removes the fixed source-row Möbius gauge, exposes the two-affine target signs, and develops pair, star, Perron, and operator-walk certificates [4]. It also proves a decisive negative fact: on a same-residual collision, the two target signs freeze. Therefore the same-residual operator has no cancellation coming from those signs alone.

The next question is not whether one can rename the unequal-residual terms as a signed perturbation. That is immediate. The useful question is *where the first genuinely new sign survives after the full operator product is expanded.* A one-edge calculation does not answer it, because operator moments join several collision edges through common source rows, and the complete native key can change from edge to edge.

We write

$$R := H_{\mathrm{res}}, \qquad E := H_{\mathrm{erase}}, \qquad H = R + E.$$

This notation is used only after the residual erasure map has been defined; it must not be confused with the represented carrier or with that map. The two defects act on the same active row space. They are separated by residual labels at the atomic output level, but they are not automatically orthogonal as matrices on the source-row space.

1.1 Main algebraic results

The first result is a key-local decomposition. For every complete native key i, the fixed-(i, m) singleton property gives one normalized amplitude $f_i(m)$, or zero, for each source row. The residual label $s_i(m)$ partitions the active rows at that key. We construct

$$R = \sum_i R_i, \qquad E = \sum_i E_i,$$

where R_i is block diagonal and E_i is off block with respect to that partition. It follows that

$$\mathrm{tr}(E_i P(R_i)) = 0$$

for every polynomial P. The first mixed trace is therefore

$$\mathrm{tr}(ER) = \sum_{i \neq j} \mathrm{tr}(E_i R_j).$$

Each term uses two rows and two distinct complete native keys. Its four literal amplitudes form a rectangle, and their product is invariant under both row phases and output-key phases. We call this product the *cross-key holonomy.*

The second result organizes every noncommutative mixed moment. If

$$\mathrm{tr}(R + tE)^k = \sum_{b=0}^{k} t^b S_{k,b},$$

then, for $1 \le b \le k$,

$$S_{k,b} = \frac{k}{b} \sum_{\substack{a_1 + \cdots + a_b = k - b \\ a_\nu \ge 0}} \mathrm{tr}(ER^{a_1} ER^{a_2} \cdots ER^{a_b}).$$

No commutativity is assumed. This is a marked-erasure counting identity. It gives $S_{k,1} = k\,\mathrm{tr}(R^{k-1}E)$ and isolates the first mixed signed sector.

The third result evaluates the signs in every decorated closed walk. An erasure edge has two unequal residual endpoints. Form the multigraph whose vertices are the integer residual labels and whose edges are these endpoint pairs. The complete product of target Möbius signs is

$$\prod_{\deg_E(s)\ \text{odd}} \mu(s).$$

All source-row gauges and all same-residual signs have disappeared exactly. Thus a residual Eulerian subgraph forces sign $+1$, while a nonempty residual boundary is the only place where the target signs can remain algebraically nontrivial.

Finally, the complete exactly-two-erasure sector in every even trace is nonnegative:

$$S_{2q,2} \geq 0.$$

This is not a heuristic positivity statement about individual decorated walks. It follows from the exact formula

$$S_{2q,2} = q \sum_{\alpha,\beta} |E_{\alpha\beta}|^2 \frac{\lambda_\alpha^{2q-1} - \lambda_\beta^{2q-1}}{\lambda_\alpha - \lambda_\beta},$$

where λ_α are the eigenvalues of R and the diagonal quotient has its derivative value. The divided difference is nonnegative because x^{2q-1} is increasing on the real line.

1.2 What these results do and do not change

The paper advances the operator interface in three ways. It shows that the first mixed second moment is necessarily cross-key; it reduces every closed walk sign to a residual boundary; and it proves that the full quadratic erasure response in an even moment cannot be negative. These facts remove several possible but incorrect proof shortcuts.

They do not estimate the resulting arithmetic sums. In particular, this paper proves none of the following:

$$\operatorname{tr}(R^{2q-1}E) = o(\text{absolute majorant}), \qquad \|R + E\| = X^{o(1)}, \qquad \|C_N\| = o(1).$$

It also proves no missing-mass saving, represented lower frame, parity estimate, prime-pair theorem, or twin-prime conclusion. The fixed shift is never averaged and is not silently specialized to 2.

1.3 Organization

Section 2 freezes the complete physical carrier. Section 3 proves the key-local diagonal/off-block decomposition. Section 4 gives the cyclic sector formula. Section 5 proves the residual-boundary sign theorem. Section 6 derives the exact two-key rectangle and its direct-cell form. Section 7 proves positivity of the quadratic even-moment response. Section 8 separates local compensation from global trace cancellation. Countermodels and the conditional arithmetic ledger follow, and the final section records L0, L1, L2, and endpoint boundaries.

2 The complete fixed-shift interface

We restate the part of the TPC–61/TPC–68 interface needed below. The purpose is to ensure that every later matrix entry is attached to an actual physical atom and a complete output key.

2.1 Atoms and recovered arithmetic labels

Fix a scale X, a nonzero integer h_0, all direct sides and external blocks in the declared scope, all source and opposite-row blocks, residue and orbit classes, masks, profiles, endpoint restrictions, and the very-high cutoff before choosing a coefficient vector. Let $\mathcal{W}_M$ be the resulting complete set of nonzero missing-high atoms and let

$$\mathcal{K}_X = \ell^2(\mathcal{M}_X)$$

be the full declared source-row space.

Every $w \in \mathcal{W}_M$ canonically recovers

$$m(w) = d_{m(w)}\ell_{m(w)}, \qquad j(w), \qquad n(w) = m(w)j(w) + h_0, \tag{1}$$

its largest prime $p(w) > y$, its cofactor

$$r(w) = \frac{n(w)}{p(w)},$$

and its residual label

$$s(w) = d_{m(w)} r(w). \tag{2}$$

The physical support is squarefree and gives

$$\mu(s(w)) = \mu(d_{m(w)})\mu(r(w)). \tag{3}$$

The complete native output key is denoted by $i(w)$. It includes the direct side, external block, opposite row, γ-coordinate, physical orbit time j, and every remaining label that is orthogonal in the native range. In particular,

$$i(w) = i(w') \quad \Longrightarrow \quad j(w) = j(w'), \tag{4}$$

but equality of i does not force equality of s. The critical-cutoff singleton property states

$$\#\{w \in \mathcal{W}_M : (i(w), m(w)) = (i, m)\} \leq 1. \tag{5}$$

It does not say that one native key supports only one row, one prime, or one residual label.

The literal atom driven by a row coefficient ζ_m is

$$\beta_w \zeta_{m(w)}.$$

The fixed nonzero coefficient β_w retains every physical source shape, mask, multiplier, opposite-row factor, output phase, profile, and terminal weight. No atomwise phase is introduced after the carrier is formed.

2.2 Atomic, residual, and native maps

Absorb the physical high character in the TPC–61 convention by putting

$$c_w = -\mu(s(w))\beta_w, \qquad (L_X\zeta)_w = c_w \zeta_{m(w)}. \tag{6}$$

Define

$$(T_X\zeta)_{i,s} := \sum_{\substack{w \in \mathcal{W}_M \\ i(w)=i,\ s(w)=s}} c_w \zeta_{m(w)}, \tag{7}$$

$$(\mathsf{E}z)_i := \sum_s z_{i,s}. \tag{8}$$

Then the literal missing-native map is

$$\boxed{M_X = \mathsf{E}T_X.} \tag{9}$$

Only s is erased. Applying a second Möbius sign after (8) would change the physical map.

For every declared row put

$$W_m := W_M(m) := \sum_{\substack{w \in \mathcal{W}_M \\ m(w)=m}} |\beta_w|^2, \qquad D_M := \operatorname{diag}(W_m)_{m \in \mathcal{M}_X}. \tag{10}$$

This is raw atomic mass, not an average over output fibers. Moreover

$$D_M = L_X^* L_X.$$

The active missing-row space is

$$\mathcal{M}_M := \{m \in \mathcal{M}_X : W_M(m) > 0\}, \qquad S_M := \operatorname{Ran} D_M = (\operatorname{Ker} D_M)^\perp = \ell^2(\mathcal{M}_M). \tag{11}$$

Rows in $\operatorname{Ker} D_M$ remain present in the full coefficient space and in the represented map. Only the normalized missing Gram is restricted to S_M.

Assume $S_M \neq \{0\}$, and define

$$F := M_X D_M^{\dagger/2}|_{S_M}, \qquad F_{\mathrm{res}} := T_X D_M^{\dagger/2}|_{S_M}. \tag{12}$$

TPC–61 gives Hermitian zero-diagonal matrices

$$R := H_{\mathrm{res}}, \qquad E := H_{\mathrm{erase}}$$

such that

$$F_{\mathrm{res}}^* F_{\mathrm{res}} = I + R, \tag{13}$$
$$F^* F = I + R + E. \tag{14}$$

Thus R contains common-(i,s) collisions already visible before erasure, whereas E contains exactly the unequal-s cross terms created at a common i by E.

2.3 The source-row gauge

On the squarefree active support, let

$$U_d = \mathrm{diag}(\mu(d_m))_{m \in \mathcal{M}_M}. \tag{15}$$

It is a self-adjoint unitary. Conjugating the normalized Gram by U_d does not change its spectrum, norms, traces of powers, or collision graph. The signed amplitude of an atom in the gauged normalized incidence is

$$f_i(m) := \frac{\mu(n(w))\beta_w}{\sqrt{W_m}} \quad \text{if } w \text{ is the unique atom with } (i(w), m(w)) = (i, m), \tag{16}$$

and $f_i(m) = 0$ when no such atom exists. Indeed,

$$\mu(d_m)c_w = -\mu(d_m)\mu(s(w))\beta_w = -\mu(r(w))\beta_w = \mu(n(w))\beta_w.$$

Consequently, for $m \neq n$, the gauged full defect has entry

$$(U_d(R+E)U_d)_{mn} = \sum_i \overline{f_i(m)} f_i(n). \tag{17}$$

Every summand uses one complete native key. In the remainder we work in these gauged coordinates and suppress U_d from the notation.

2.4 Direct-cell specialization

Only in the identical-support direct-cell specialization, a displayed left-side atom may be written $w = (m, j, \gamma)$ with

$$\beta_{m,j,\gamma} = a_m(j) K_{m\gamma} q_\gamma(j), \qquad K_{m\gamma} \in \{0, 1\}. \tag{18}$$

For two rows define the literal orthogonal output sum

$$\Gamma_{mn}(j) = \sum_\gamma K_{m\gamma} K_{n\gamma} |q_\gamma(j)|^2. \tag{19}$$

The γ-coordinate has not been deleted in this formula; it has been summed exactly in the Gram over orthogonal output coordinates. Outside the declared specialization, one must keep the full atom pair sum or place the complete γ-dependent first-failure indicator inside (19).

The opposite direct side is orthogonal at the output level. Hence its Gram adds to the displayed-side Gram. This eliminates $F_L^* F_R$, but it does not imply

$$H^L H^R = 0.$$

An operator walk may therefore use a different side, block, γ, or time on each edge.

2.5 Six-item physical audit

The interface satisfies:

(L1.1) every atom uses the literal coefficient $-\mu(s(w))\beta_w$;

(L1.2) every target is $m(w)j(w) + h_0$ for one fixed $h_0 \neq 0$;

(L1.3) D_M is raw atomic mass, with no fiber normalization;

(L1.4) source rows and complete native outputs are canonical, and only s is erased;

(L1.5) the carrier and all key partitions are fixed before choosing a coefficient vector or an operator extremizer; and

(L1.6) the identities in this paper have zero structural exponent cost, while every later estimate remains subject to the strict $1/400$ endpoint ledger.

3 Key-local residual blocks and erasure links

This section proves that one erasure edge cannot close against powers of the same native-key residual operator. The conclusion is purely finite-dimensional and uses the complete key rather than a reduced collision cell.

3.1 The residual partition at one native key

For a native key i, let

$$A_i := \{m \in \mathcal{M}_M : f_i(m) \neq 0\}.$$

For $m \in A_i$, let $s_i(m)$ denote the residual label of the unique atom selected by (i, m). Extend f_i by zero to all of S_M, and put

$$D_i := \operatorname{diag}\big(|f_i(m)|^2\big)_{m \in \mathcal{M}_M}. \tag{20}$$

Define two positive matrices by their entries:

$$(K_i)_{mn} := \overline{f_i(m)} f_i(n), \tag{21}$$

$$(K_i^{\mathrm{res}})_{mn} := \mathbf{1}_{\{s_i(m)=s_i(n)\}} \overline{f_i(m)} f_i(n), \tag{22}$$

where an indicator is zero if either amplitude vanishes. Equivalently, if

$$A_{i,s} := \{m \in A_i : s_i(m) = s\},$$

then K_i^{res} is the orthogonal direct sum of the rank-one Grams on the coordinate spaces $\ell^2(A_{i,s})$.

Definition 3.1 (Key-local defects). Set

$$R_i := K_i^{\mathrm{res}} - D_i, \qquad E_i := K_i - K_i^{\mathrm{res}}. \tag{23}$$

Thus R_i has the same-residual off-diagonal entries at i, and E_i has the unequal-residual entries at i. Both have zero diagonal.

Proposition 3.2 (Exact complete-key decomposition)**.** *On S_M,*

$$\boxed{\sum_i D_i = I, \qquad R = \sum_i R_i, \qquad E = \sum_i E_i.} \tag{24}$$

Every sum is over the actual complete native output keys.

Proof. For a fixed active row m, the fixed-(i, m) singleton property and (16) give

$$\sum_i |f_i(m)|^2 = \frac{1}{W_m} \sum_{\substack{w \in \mathcal{W}_M \\ m(w)=m}} |\beta_w|^2 = 1.$$

This proves $\sum_i D_i = I$. Summing K_i^{res} over i gives the normalized residual Gram $I + R$, because equality of both i and s is precisely equality of the residual output. Summing K_i gives the native Gram $I + R + E$. Subtracting the diagonal identity proves the other two formulas. $\square$

3.2 The local off-block theorem

Let

$$\mathscr{P}_i := \bigoplus_s \ell^2(A_{i,s}) \oplus \ell^2(A_i^c) \tag{25}$$

be the coordinate decomposition induced by residual labels at i. Relative to $\mathscr{P}_i$, R_i is block diagonal. The diagonal blocks of E_i, including the A_i^c-block, are zero.

Theorem 3.3 (One-key mixed traces vanish). *For every complete native key i and every integer $a \geq 0$,*

$$\boxed{\operatorname{tr}(E_i R_i^a) = 0.} \tag{26}$$

More generally,

$$\operatorname{tr}(E_i P(R_i)) = 0 \tag{27}$$

for every polynomial P.

Proof. Every power of the block-diagonal matrix R_i is block diagonal relative to $\mathscr{P}_i$. The product of the strictly off-block matrix E_i with a block-diagonal matrix has zero diagonal blocks and hence zero trace. Linearity gives the polynomial statement. The case $a = 0$ is simply $\operatorname{tr} E_i = 0$. □

Corollary 3.4 (The first mixed trace is cross-key). *One has*

$$\boxed{\operatorname{tr}(ER) = \sum_{\substack{i,j \\ i \neq j}} \operatorname{tr}(E_i R_j).} \tag{28}$$

Proof. Expand $E = \sum_i E_i$ and $R = \sum_j R_j$. The terms with $i = j$ vanish by theorem 3.3 with $a = 1$. □

Remark 3.5 (Disjoint pair lists do not imply matrix orthogonality). At one key, an atom pair cannot be simultaneously equal-residual and unequal-residual. This set-theoretic disjointness proves $\operatorname{tr}(E_i R_i) = 0$. It does not prove $\operatorname{tr}(ER) = 0$, because E_i and R_j with $i \neq j$ are matrices on the same row space and may occupy the same row-pair entry.

3.3 Higher one-erasure words

The local theorem removes only the word whose erasure edge and all residual edges carry the same native key. Indeed,

$$\operatorname{tr}(ER^{k-1}) = \sum_{i, j_1, \ldots, j_{k-1}} \operatorname{tr}(E_i R_{j_1} \cdots R_{j_{k-1}}). \tag{29}$$

The term with $j_1 = \cdots = j_{k-1} = i$ vanishes. Other key sequences need not vanish, even if some adjacent keys agree. Thus the one-erasure sector of a longer moment is a cross-key polygon rather than a one-key ladder.

Remark 3.6 (Why complete keys matter). If side, external block, opposite row, γ, or orbit time is removed from i, two different physical output coordinates can be placed in one artificial local block. The proof of (26) would then concern the compressed model, not the native Gram. Conversely, splitting one actual key into coefficient- dependent subkeys would manufacture orthogonality. Neither operation is allowed.

4 Cyclic sectors of noncommutative mixed words

We now organize all words in R and E before using their arithmetic entries. The identities in this section hold for arbitrary finite square matrices for which the displayed traces are defined.

4.1 The erasure-count generating polynomial

For integers $k \geq 1$ and $0 \leq b \leq k$, define

$$S_{k,b} := \sum_{\substack{\omega \in \{R,E\}^k \\ |\omega|_E = b}} \operatorname{tr}(H_{\omega_1} \cdots H_{\omega_k}), \qquad H_R = R, \quad H_E = E. \tag{30}$$

Equivalently,

$$\operatorname{tr}(R + tE)^k = \sum_{b=0}^{k} t^b S_{k,b}. \tag{31}$$

The order of the letters is retained in (30); no binomial commutativity is assumed.

Theorem 4.1 (Exact cyclic erasure-sector formula)**.** *For $1 \leq b \leq k$,*

$$\boxed{S_{k,b} = \frac{k}{b} \sum_{\substack{a_1 + \cdots + a_b = k-b \\ a_\nu \geq 0}} \operatorname{tr}(ER^{a_1} ER^{a_2} \cdots ER^{a_b})\,.} \tag{32}$$

Also $S_{k,0} = \operatorname{tr} R^k$.

Proof. Mark one of the b occurrences of E in every word contributing to $S_{k,b}$. This produces b marked copies of each linear word. Rotate a marked word so that its marked E is first. The numbers of consecutive R's after the successive E's form a weak composition

$$a_1 + \cdots + a_b = k - b.$$

Conversely, a weak composition and one of the k possible rotation offsets reconstruct a marked linear word. Trace cyclicity makes all rotations have the displayed trace. Therefore $bS_{k,b}$ is k times the sum over weak compositions, proving (32). The counting remains valid for periodic words because the marked occurrence and rotation offset are retained. □

Corollary 4.2 (The one- and two-erasure sectors)**.** *For $k \geq 2$,*

$$S_{k,1} = \boxed{k \operatorname{tr}(ER^{k-1})}, \tag{33}$$

$$S_{k,2} = \boxed{\frac{k}{2} \sum_{a=0}^{k-2} \operatorname{tr}(ER^a ER^{k-2-a}).} \tag{34}$$

The first two even expansions are

$$\operatorname{tr}(R + E)^2 = \operatorname{tr} R^2 + 2 \operatorname{tr}(RE) + \operatorname{tr} E^2, \tag{35}$$

$$\operatorname{tr}(R + E)^4 = \operatorname{tr} R^4 + 4 \operatorname{tr}(R^3 E) + 4 \operatorname{tr}(R^2 E^2) + 2 \operatorname{tr}(RERE) + 4 \operatorname{tr}(RE^3) + \operatorname{tr} E^4. \tag{36}$$

The two distinct two-erasure necklaces in (36) must not be merged before their traces are evaluated.

4.2 Key expansion of the one-erasure sector

Combining proposition 3.2 and corollary 4.2 gives

$$S_{k,1} = k \sum_{i, j_1, \ldots, j_{k-1}} \operatorname{tr}(E_i R_{j_1} \cdots R_{j_{k-1}}). \tag{37}$$

By theorem 3.3, the fully local term

$$\operatorname{tr}(E_i R_i^{k-1})$$

vanishes. Hence each surviving decorated contribution changes complete native key at least once. At $k = 2$ this forces two distinct keys exactly. For $k > 2$, it produces cross-key polygons with one off-residual edge.

Remark 4.3 (Sector control versus moment control). A bound for $S_{2q,1}$ alone is not a bound for $\operatorname{tr}(R{+}E)^{2q}$. The zero-erasure sector, the nonnegative quadratic sector proved in section 7, and all higher sectors remain. Conversely, a bad estimate for one sector does not prove that the full moment is large, because different sectors can cancel after t is set to one. A valid moment argument must state which sectors are bounded absolutely and which are combined with their literal signs.

4.3 Decorated key words

For later use, a decorated realization of a word consists of:

(i) a row path $m_0, m_1, \dots, m_k$;

(ii) a complete native key i_ℓ on each edge;

(iii) the unique atoms $w_{\ell,-}$, $w_{\ell,+}$ at $(i_\ell, m_{\ell-1})$ and (i_ℓ, m_ℓ); and

(iv) the condition
$$s(w_{\ell,-}) = s(w_{\ell,+}) \quad \Longleftrightarrow \quad \omega_\ell = R.$$

The row path joins edges, but it does not identify their keys, atoms, times, residual labels, sides, or γ-coordinates. This distinction is responsible for both the cross-key rectangles below and the absence of a general telescope for literal complex weights.

5 Residual-boundary signs on decorated walks

TPC–68 proves that target signs freeze on a same-residual edge [4]. We strengthen the bookkeeping from a pure residual walk to an arbitrary mixed word and identify the exact surviving sign.

5.1 The edge identity

Consider an oriented decorated edge
$$\xi = (w_-, w_+)$$
from row m to row n, with common complete native key. Abbreviate
$$s_-(\xi) = s(w_-), \qquad s_+(\xi) = s(w_+).$$
The gauged target sign is
$$\varepsilon(\xi) := \mu(n(w_-))\mu(n(w_+)).$$

Lemma 5.1 (Source and residual factorization of one edge)**.** *On the squarefree carrier,*
$$\boxed{\varepsilon(\xi) = \mu(d_m)\mu(d_n)\mu(s_-(\xi))\mu(s_+(\xi)).} \tag{38}$$
If ξ is an R-edge, this reduces to
$$\varepsilon(\xi) = \mu(d_m)\mu(d_n). \tag{39}$$

Proof. Since $n(w) = p(w)r(w)$ is squarefree and $p(w)$ is prime,
$$\mu(n(w)) = -\mu(r(w)) = -\mu(d_{m(w)})\mu(s(w)).$$
Multiplying the two endpoint identities proves (38). On an R-edge the two residual labels agree, so their Möbius factors square to one. □

5.2 Open paths and closed walks

Theorem 5.2 (Exact residual-boundary sign)**.** *Let*
$$m_0, m_1, \dots, m_k$$
be a decorated row path with edge types $\omega_\ell \in \{R, E\}$. Then
$$\boxed{\prod_{\ell=1}^{k} \varepsilon(\xi_\ell) = \mu(d_{m_0})\mu(d_{m_k}) \prod_{\ell:\omega_\ell = E} \mu(s_-(\xi_\ell))\mu(s_+(\xi_\ell)).} \tag{40}$$
If $m_k = m_0$, the source-row boundary disappears:
$$\boxed{\prod_{\ell=1}^{k} \varepsilon(\xi_\ell) = \prod_{\ell:\omega_\ell = E} \mu(s_-(\xi_\ell))\mu(s_+(\xi_\ell)).} \tag{41}$$

Proof. Apply lemma 5.1 to every edge. Each intermediate row gauge $\mu(d_{m_\ell})$ occurs twice and squares to one. Only the endpoint row gauges remain on an open path. Every R-edge residual pair also squares to one, leaving exactly (40). On a closed path the two endpoint gauges agree and square to one. □

For a closed decorated walk, build a residual multigraph $\mathscr{G}_E$ as follows. Its vertices are the integer residual labels appearing on E-edge endpoints, and every E-edge contributes the unordered residual edge

$$\{s_-(\xi), s_+(\xi)\}.$$

The two endpoints are distinct by definition, although parallel edges are allowed. Let $\deg_E(s)$ be the degree of residual vertex s, counted with multiplicity.

Corollary 5.3 (Odd residual boundary)**.** *The closed-walk target sign is*

$$\boxed{\prod_s \mu(s)^{\deg_E(s)} = \prod_{\deg_E(s)\ \text{odd}} \mu(s).} \tag{42}$$

In particular, if $\mathscr{G}_E$ is Eulerian, the target sign is forced to be $+1$.

Proof. Group the factors in (41) by residual label and use $\mu(s)^2 = 1$. □

Remark 5.4 (Edge-count parity is not residual parity)*.* The number of E-edges does not determine the sign. Two edges $(a,b),(b,c)$ leave $\mu(a)\mu(c)$, while three edges forming the residual triangle $(a,b),(b,c),(c,a)$ force sign $+1$. A nonempty odd boundary means only that the sign is not algebraically forced by the interface; its value on one actual carrier can still happen to equal $+1$.

5.3 The literal closed-walk expansion

For a closed decorated word of length k, the product of the normalizing denominators is

$$\prod_{\ell=0}^{k-1} W_{m_\ell}.$$

Combining this with theorem 5.2 gives

$$\boxed{\operatorname{tr}(H_{\omega_1}\cdots H_{\omega_k}) = \sum_{\substack{m_k=m_0\\ \xi_\ell\in\mathcal{C}^{\omega_\ell}_{m_{\ell-1}m_\ell}}} \left[\prod_{\ell:\omega_\ell=E} \mu(s_-(\xi_\ell))\mu(s_+(\xi_\ell))\right] \times \frac{\displaystyle\prod_{\ell=1}^{k} \overline{\beta_{w_{\ell,-}}}\beta_{w_{\ell,+}}}{\displaystyle\prod_{\ell=0}^{k-1} W_{m_\ell}}.} \tag{43}$$

Here $\mathcal{C}^R_{mn}$ and $\mathcal{C}^E_{mn}$ are the complete equal- and unequal-residual native collision lists. The sum retains the native key, side, block, opposite row, γ, time, mask, and every literal coefficient on every edge.

The complex phase product in (43) does not generally telescope. It becomes positive on an exactly atom-consistent backtracking decoration, but adjacent matrix edges share only a source-row index, not necessarily an atom. Treating every complex weight as a vertex gauge would therefore change the sum.

Corollary 5.5 (One-erasure sign)**.** *Every decorated term in the one-erasure sector has precisely the residual sign*

$$\mu(s_-(\xi_E))\mu(s_+(\xi_E)), \tag{44}$$

where ξ_E is its unique E-edge. No same-residual edge contributes a target Möbius sign after the closed product is formed.

6 Two-key rectangles and cross-key holonomy

The shortest mixed closed word is ER. By corollary 3.4, it necessarily uses two distinct complete native keys. We now write its four corners explicitly.

6.1 Exact rectangle formula

For keys i, j and rows m, n, define

$$\mathcal{H}_{i,j}(m,n) := \overline{f_i(m)} f_i(n) \overline{f_j(n)} f_j(m). \tag{45}$$

It vanishes when one of the four key-row corners is absent. Throughout this section, a residual-relation indicator is also defined to be zero if either amplitude named by that indicator vanishes.

Theorem 6.1 (Cross-key rectangle identity). *For $i \neq j$,*

$$\boxed{\operatorname{tr}(E_i R_j) = 2 \operatorname{Re} \sum_{m<n} \mathbf{1}_{\{s_i(m) \neq s_i(n)\}} \mathbf{1}_{\{s_j(m)=s_j(n)\}} \mathcal{H}_{i,j}(m,n).} \tag{46}$$

Consequently,

$$\boxed{\operatorname{tr}(ER) = 2 \operatorname{Re} \sum_{\substack{i \neq j \\ m<n}} \mathbf{1}_{\{s_i(m) \neq s_i(n)\}} \mathbf{1}_{\{s_j(m)=s_j(n)\}} \mathcal{H}_{i,j}(m,n).} \tag{47}$$

Proof. By definition,

$$\operatorname{tr}(E_i R_j) = \sum_{m,n} (E_i)_{mn} (R_j)_{nm}.$$

The diagonal terms vanish. For $m \neq n$, substitute

$$(E_i)_{mn} = \mathbf{1}_{\{s_i(m) \neq s_i(n)\}} \overline{f_i(m)} f_i(n)$$

and

$$(R_j)_{nm} = \mathbf{1}_{\{s_j(n)=s_j(m)\}} \overline{f_j(n)} f_j(m).$$

The ordered terms (m, n) and (n, m) are complex conjugates, giving (46). Sum over $i \neq j$ and use (28). $\square$

The key i carries the unequal-residual edge and j carries the same-residual edge. The roles are not interchangeable in (47); the term with (j, i) is present only if the residual relations are reversed at those keys.

6.2 Gauge invariance

Proposition 6.2 (Rectangle holonomy is gauge invariant). *Let $u_m, v_i \in \mathbb{C}$ have modulus one and replace*

$$f_i(m) \longmapsto u_m v_i f_i(m).$$

Then every $\mathcal{H}_{i,j}(m,n)$ is unchanged.

Proof. In (45), the row phases occur as

$$\overline{u_m} u_n \overline{u_n} u_m = 1$$

and the key phases as

$$\overline{v_i} v_i \overline{v_j} v_j = 1.$$

$\square$

Thus the rectangle phase cannot be removed by moving a unit phase between source rows and outputs. It is a genuine four-corner quantity. Support and coefficient magnitudes do not determine it.

6.3 The residual sign on a rectangle

Let $\beta_{i,m}$ be the literal coefficient of the unique atom at (i, m), extended by zero. The row-gauged amplitude is

$$f_i(m) = \mu(n_{i,m})\frac{\beta_{i,m}}{\sqrt{W_m}}, \qquad n_{i,m} = m\,j(i) + h_0.$$

Proposition 6.3 (Four target signs reduce to one erasure pair)**.** *On a rectangle contributing to* (46)*,*

$$\boxed{\mathcal{H}_{i,j}(m,n) = \mu(s_i(m))\mu(s_i(n))\frac{\overline{\beta_{i,m}}\beta_{i,n}\overline{\beta_{j,n}}\beta_{j,m}}{W_mW_n}.} \tag{48}$$

Proof. The product of the four target signs is

$$\mu(n_{i,m})\mu(n_{i,n})\mu(n_{j,n})\mu(n_{j,m}).$$

Use

$$\mu(n_{k,m}) = -\mu(d_m)\mu(s_k(m)).$$

The four minus signs cancel, and each source-row sign occurs twice. At the same-residual key j, $\mu(s_j(m))\mu(s_j(n)) = 1$. The two residual signs at the erasure key i remain, proving the formula. □

Equation (48) is the minimal instance of the residual-boundary theorem. It also shows why a one-key estimate cannot see the first mixed response: the sign-bearing unequal-residual edge and its same-residual return occupy different complete native keys.

6.4 The direct-cell two-time rectangle

In the identical-support direct-cell specialization, let c collect a fixed side and external cell, and let

$$\mathfrak{m}^c_m(t)$$

be the actual missing-time indicator. Put

$$\chi^{mn}_E(t) = \mathbf{1}_{\{s_m(t)\neq s_n(t)\}}, \qquad \chi^{mn}_R(t) = \mathbf{1}_{\{s_m(t)= s_n(t)\}}.$$

After summing the two independent orthogonal γ-coordinates exactly, (47) becomes

$$\boxed{\begin{aligned}\operatorname{tr}(ER) = 2\operatorname{Re}\sum_{m<n}\frac{1}{W_mW_n}\sum_{\substack{c,t\\ c',u}} &\mathfrak{m}^c_m(t)\mathfrak{m}^c_n(t)\mathfrak{m}^{c'}_n(u)\mathfrak{m}^{c'}_m(u)\\ &\times \chi^{mn}_E(t)\chi^{mn}_R(u)\mu(s_m(t))\mu(s_n(t))\\ &\times \overline{a^c_m(t)}a^c_n(t)\overline{a^{c'}_n(u)}a^{c'}_m(u)\Gamma^c_{mn}(t)\Gamma^{c'}_{nm}(u).\end{aligned}} \tag{49}$$

The two times t and u, two sides/cells c, c', and the two γ-sums are independent. Formula (49) is invalid outside the declared specialization unless the full γ-dependent first-failure indicators are restored.

6.5 Two determinant edges

For the i-edge at time $t = j(i)$, write

$$p_{i,m}r_{i,m} = mt + h_0, \qquad p_{i,n}r_{i,n} = nt + h_0.$$

Then

$$np_{i,m}r_{i,m} - mp_{i,n}r_{i,n} = h_0(n-m). \tag{50}$$

The same identity holds at key j, generally with another time. The rectangle conditions are

$$d_mr_{i,m} \neq d_nr_{i,n}, \tag{51}$$
$$d_mr_{j,m} = d_nr_{j,n}. \tag{52}$$

Thus the first mixed trace is a pair of determinant solutions sharing two source rows, with unequal residual labels on one horizontal edge and equal residual labels on the other. The determinant identities are exact reindexings; they do not pair the signs in (48).

7 The nonnegative quadratic response of even moments

The residual-boundary formula shows that individual two-erasure decorations can retain nontrivial signs. Nevertheless, after all words with exactly two E's are assembled, the even-moment coefficient has a definite sign. This is a global matrix identity, not an arithmetic estimate.

7.1 Divided-difference formula

Theorem 7.1 (Nonnegative two-erasure sector)**.** *Let R and E be finite Hermitian matrices. For every integer $q \geq 1$,*

$$\boxed{S_{2q,2} \geq 0.} \tag{53}$$

More precisely, choose an orthonormal eigenbasis (e_α) with

$$Re_\alpha = \lambda_\alpha e_\alpha, \qquad P_\alpha = e_\alpha e_\alpha^*, \qquad R = \sum_\alpha \lambda_\alpha P_\alpha,$$

where equal eigenvalues may occur at different indices, and put $E_{\alpha\beta} = \langle Ee_\beta, e_\alpha\rangle$. Then

$$\boxed{S_{2q,2} = q \sum_{\alpha,\beta} |E_{\alpha\beta}|^2 \Phi_q(\lambda_\alpha, \lambda_\beta),} \tag{54}$$

where

$$\Phi_q(x,y) := \sum_{a=0}^{2q-2} x^{2q-2-a} y^a = \begin{cases} \dfrac{x^{2q-1} - y^{2q-1}}{x-y}, & x \neq y, \\ (2q-1)x^{2q-2}, & x = y. \end{cases} \tag{55}$$

Proof. By (34),

$$S_{2q,2} = q \sum_{a=0}^{2q-2} \operatorname{tr}(ER^a E R^{2q-2-a}).$$

In an eigenbasis of R,

$$\operatorname{tr}(ER^a E R^{2q-2-a}) = \sum_{\alpha,\beta} |E_{\alpha\beta}|^2 \lambda_\beta^a \lambda_\alpha^{2q-2-a}.$$

Summing over a gives (54). The function

$$g_q(x) = x^{2q-1}$$

is increasing on $\mathbb{R}$, because

$$g_q'(x) = (2q-1)x^{2q-2} \geq 0.$$

Therefore every off-diagonal divided difference in (55) is nonnegative, and the diagonal value is also nonnegative. Each summand in (54) is nonnegative. $\square$

Corollary 7.2 (Quantitative upper bound and equality cases)**.** *One has*

$$0 \leq S_{2q,2} \leq \binom{2q}{2} \|R\|^{2q-2} \|E\|_{\mathrm{HS}}^2. \tag{56}$$

For $q = 1$,

$$S_{2,2} = \|E\|_{\mathrm{HS}}^2.$$

For $q \geq 2$, equality $S_{2q,2} = 0$ holds if and only if

$$E = P_{\operatorname{Ker} R} E P_{\operatorname{Ker} R}. \tag{57}$$

Proof. From the finite sum in (55),

$$|\Phi_q(x,y)| \leq (2q-1) \max\{|x|, |y|\}^{2q-2}.$$

Insert this into (54) and use $q(2q-1) = \binom{2q}{2}$. When $q = 1$, $\Phi_1 = 1$. For $q \geq 2$, the odd power x^{2q-1} is strictly increasing, so $\Phi_q(x,y) > 0$ unless $x = y = 0$. Hence the sum vanishes exactly when every nonzero matrix entry of E lies inside the zero eigenspace of R. $\square$

7.2 Response interpretation

Put

$$\mathcal{F}_q(t) := \operatorname{tr}(R + tE)^{2q}. \tag{58}$$

Then

$$\mathcal{F}_q'(0) = S_{2q,1} = 2q \operatorname{tr}(ER^{2q-1}), \tag{59}$$

$$\frac{1}{2}\mathcal{F}_q''(0) = S_{2q,2} \geq 0. \tag{60}$$

Applying the theorem with R replaced by $R + tE$ shows that $\mathcal{F}_q''(t) \geq 0$ for every real t. Thus $\mathcal{F}_q$ is convex. This is also consistent with the convexity of the $2q$-th power of the Schatten $2q$-norm [1, 2].

The conclusion is deliberately limited:

- the linear response can have either sign;
- the complete quadratic response cannot be negative;
- sectors of order three and above can again be signed; and
- convexity in t does not by itself compare $\mathcal{F}_q(1)$ with $\mathcal{F}_q(0)$, because the initial derivative may be negative.

7.3 Individual words need not be positive

The positivity in theorem 7.1 occurs only after every two-erasure word is assembled with its correct cyclic multiplicity. For example, take

$$R = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\operatorname{tr}(RERE) = -2, \qquad \operatorname{tr}(R^2E^2) = 2.$$

At fourth order,

$$S_{4,2} = 4\operatorname{tr}(R^2E^2) + 2\operatorname{tr}(RERE) = 4 > 0.$$

Taking the alternating word alone would therefore give the wrong sign for the full quadratic sector.

7.4 Arithmetic meaning

At the literal level, (53) says that all two-erasure residual-boundary signs and all complex coefficient phases, after being summed over every word position and every decorated walk, form a nonnegative total. It supplies no smallness. Indeed, the upper bound in (56) can be large. The theorem closes only a proposed mechanism: the complete quadratic sector cannot be invoked as a negative correction that cancels the zero-erasure moment.

The first potentially favorable perturbative term is therefore the cross-key linear response

$$S_{2q,1} = 2q \operatorname{tr}(ER^{2q-1}),$$

whose residual-boundary sign remains an L2 arithmetic problem.

8 Local star compensation and global trace

Operator moments permit cancellation between closed walks starting at different rows. The star certificate of TPC–68 instead tests each row locally. We record the exact relation between the two viewpoints.

8.1 Rowwise compensation identity

For an active row m, let

$$r_m = (R_{mn})_{n\neq m}, \qquad e_m = (E_{mn})_{n\neq m}$$

be its residual and erasure stars. Define

$$s_m(R) := \|r_m\|_2, \qquad s_m(E) := \|e_m\|_2, \qquad s_m(H) := \|r_m + e_m\|_2, \tag{61}$$

$$c_m := 2\,\mathrm{Re}\sum_{n\neq m} R_{mn}\overline{E_{mn}}. \tag{62}$$

Proposition 8.1 (Exact local compensation)**.** *For every active row,*

$$\boxed{s_m(H)^2 = s_m(R)^2 + s_m(E)^2 + c_m.} \tag{63}$$

Moreover,

$$\big(s_m(R) - s_m(E)\big)^2 \le s_m(H)^2 \le \big(s_m(R) + s_m(E)\big)^2. \tag{64}$$

The global mixed trace is the sum of local compensations:

$$\boxed{2\,\mathrm{tr}(RE) = \sum_m c_m.} \tag{65}$$

Proof. Expand $\|r_m + e_m\|_2^2$ and apply Cauchy–Schwarz to its cross term. Since R, E are Hermitian,

$$\mathrm{tr}(RE) = \sum_{m,n} R_{mn}E_{nm} = \sum_m \mathrm{Re}\sum_{n\neq m} R_{mn}\overline{E_{mn}}.$$

This proves the final identity. □

Thus a small $s_m(H)$ in a row where both component stars are large requires almost extremal negative alignment:

$$c_m \approx -2s_m(R)s_m(E).$$

A cancellation statement for the scalar sum $\sum_m c_m$ does not imply this rowwise alignment.

8.2 What is needed for a star route

Let

$$s_*(H) := \max_m s_m(H)$$

and let Δ_H be the maximum degree of the actual nonzero collision graph. TPC–68 proves

$$s_*(H) \le \|H\| \le \sqrt{\Delta_H}\, s_*(H). \tag{66}$$

Consequently, a target $s_*(H) \le \eta$ requires the uniform inequalities

$$c_m \le \eta^2 - s_m(R)^2 - s_m(E)^2 \qquad \text{for every active } m. \tag{67}$$

An average over rows, a density-one set of rows, or a cancellation between positive c_m's and negative c_m's does not prove (67).

The rectangle formula refines the local cross term:

$$\frac{c_m}{2} = \mathrm{Re}\sum_{\substack{i\neq j\\ n\neq m}} \mathbf{1}_{\{s_i(m)\neq s_i(n)\}}\mathbf{1}_{\{s_j(m)=s_j(n)\}}\mathcal{H}_{i,j}(m,n). \tag{68}$$

Thus a local star proof requires a cross-key rectangle estimate uniformly in the base row m, not merely the complete global sum in (47).

8.3 Why a normalized global trace is insufficient

For a zero-diagonal Hermitian matrix,

$$\operatorname{tr} H^2 = \sum_m s_m(H)^2. \tag{69}$$

The unnormalized trace is a valid upper certificate:

$$\|H\|^2 \le \operatorname{tr} H^2.$$

However, in a row space of growing dimension r, an estimate for the average $r^{-1} \operatorname{tr} H^2$ does not bound $\|H\|$. One nonzero two-row block can have norm one while $r^{-1} \operatorname{tr} H^2 \to 0$ after isolated rows are adjoined.

Similarly, $\operatorname{tr}(RE) = 0$ says only that the local compensations sum to zero. It permits one component with coherent reinforcement and another with exact cancellation. A countermodel is given in example 9.4.

For higher even moments the full unnormalized quantity

$$\left(\operatorname{tr} H^{2q}\right)^{1/(2q)}$$

does bound $\|H\|$. But a fixed normalized moment or one isolated sector need not have the correct scale. If $q = q_X$ grows, all constants in the literal decorated-walk estimate must be uniform in both X and q_X.

9 Separating countermodels

The following finite-dimensional examples test the exact logical boundary of the preceding identities. They are not asserted to be realizations of the full arithmetic carrier.

Example 9.1 (One key can contain both defects but no local mixed trace)**.** *Take one native key i, three rows, amplitudes*

$$f_i = (1, 1, 1),$$

and residual classes

$$A_{i,a} = \{1, 2\}, \qquad A_{i,b} = \{3\}.$$

Then R_i contains the edge $1 \leftrightarrow 2$, while E_i contains the edges $1 \leftrightarrow 3$ and $2 \leftrightarrow 3$. Both matrices are nonzero. Nevertheless,

$$\operatorname{tr}(E_i R_i^a) = 0$$

for every $a \ge 0$, because R_i^a remains inside the two residual blocks and E_i crosses them. Thus the local vanishing theorem is not the trivial assertion that one of the two matrices is zero.

Example 9.2 (Two keys create the first mixed rectangle)**.** *Take two rows and two native keys i, j, with*

$$f_i = 2^{-1/2}(1, 1), \qquad f_j = 2^{-1/2}(1, 1).$$

At i, give the two rows different residual labels; at j, give them the same residual label. Each normalized row has total squared mass one. The only nonzero off-diagonal entries are

$$(E_i)_{12} = \frac{1}{2}, \qquad (R_j)_{12} = \frac{1}{2}.$$

Hence

$$\operatorname{tr}(E_i R_j) = \frac{1}{2} \ne 0.$$

The full Gram is positive because it is assembled from the two key incidences. This disproves any inference from one-key vanishing to global Hilbert–Schmidt orthogonality.

Example 9.3 (Same support and magnitudes, opposite holonomy). *Keep the preceding support and residual partitions, but put*

$$f_i = 2^{-1/2}(1,1), \qquad f_j = 2^{-1/2}(1,e^{\mathrm{i}\theta}).$$

Then

$$\operatorname{tr}(E_i R_j) = \frac{1}{2}\cos\theta.$$

At $\theta = 0$, R_j and E_i reinforce and the full defect has off-diagonal entry one. At $\theta = \pi$, they cancel and the full defect vanishes. The two models have identical key support, residual partitions, atomic magnitudes, and row masses. Their difference is exactly the rectangle holonomy.

Example 9.4 (Zero global cross trace with a large local star). *Take the orthogonal direct sum of two copies of example 9.3: use $\theta = 0$ on rows $1,2$ and $\theta = \pi$ on rows $3,4$. The two cross traces are $1/2$ and $-1/2$, so*

$$\operatorname{tr}(ER) = 0.$$

On the first component the full defect has off-diagonal entry one, while on the second it vanishes. Therefore

$$s_1(H) = s_2(H) = 1, \qquad \|H\| = 1.$$

Global rectangle cancellation gives no uniform local compensation.

Example 9.5 (Erasure-edge parity is not sign parity). *Assign residual endpoint pairs*

$$(a,b),(b,c)$$

to two erasure edges. Their sign is $\mu(a)\mu(c)$, not identically one. By contrast, the three erasure edges

$$(a,b),(b,c),(c,a)$$

have sign one. Thus evenness of the number of erasure edges is neither necessary nor sufficient for an algebraically frozen target sign.

Example 9.6 (Averaged second trace can miss a fixed obstruction). *Let H_r be the $r \times r$ Hermitian matrix whose only nonzero block is*

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

on the first two rows. Then

$$\|H_r\| = 1, \qquad \operatorname{tr} H_r^2 = 2, \qquad \frac{1}{r}\operatorname{tr} H_r^2 \longrightarrow 0.$$

An averaged row or normalized moment estimate therefore cannot replace an unnormalized operator certificate.

Remark 9.7 (Scope of the countermodels). The examples prove non-implications for the finite-dimensional interface. They neither show that the actual fixed-shift carrier realizes the bad phases nor show that it avoids them. Deciding that question requires the literal arithmetic weights and is L2.

10 Conditional arithmetic targets and reassembly ledger

The preceding algebra identifies the first places where unequal-residual Möbius signs can survive. This section states what an arithmetic estimate would have to control and how such an estimate would enter the existing endpoint ledger.

10.1 The first rectangle target

For complete keys $i \neq j$, define the literal oriented rectangle sum

$$\begin{aligned}\mathfrak{X}_{i,j} := \sum_{m<n} & \mathbf{1}_{\{s_i(m)\neq s_i(n)\}}\mathbf{1}_{\{s_j(m)=s_j(n)\}} \\ & \times \mu(s_i(m))\mu(s_i(n))\frac{\overline{\beta_{i,m}}\beta_{i,n}\overline{\beta_{j,n}}\beta_{j,m}}{W_m W_n}.\end{aligned} \tag{70}$$

Then

$$\operatorname{tr}(ER) = 2\operatorname{Re}\sum_{i\neq j}\mathfrak{X}_{i,j}. \tag{71}$$

The notation in (70) includes only actual four-corner rectangles. A zero coefficient at any corner deletes the term. The complete key i contains its side, external block, opposite row, γ, and time; the same is true independently for j.

There are three inequivalent scopes for a possible estimate:

(i) a bound for one fixed (i, j);

(ii) a bound after summing all key pairs incident to one fixed row m;

(iii) a bound for the complete trace sum.

The first does not imply the second when the key degree grows. The third does not imply uniform local star compensation. A proof must state which scope it supplies.

10.2 Higher one-erasure targets

For $q \geq 1$, put

$$\mathfrak{L}_{2q}(X) := S_{2q,1} = 2q\operatorname{tr}(ER^{2q-1}). \tag{72}$$

By (43), this is a complete weighted sum of cross-key closed walks, each carrying one residual pair sign. A useful L2 estimate would compare

$$|\mathfrak{L}_{2q}(X)|$$

with its literal absolute majorant, with constants uniform in all physical blocks and, if $q = q_X$ grows, uniform in q_X.

Such an estimate cannot be replaced by:

- randomness of independent Möbius signs;
- an average over h;
- a density-one set of row or key pairs;
- a fixed-dimensional computation;
- the determinant bijection alone; or
- cancellation on the pure R words, whose target signs are already frozen.

10.3 A conditional full-moment certificate

Proposition 10.1 (Conditional moment-to-native certificate)**.** *Let $q = q_X \geq 1$. Suppose an L2 argument for the complete literal carrier proves*

$$\operatorname{tr}(R+E)^{2q} \leq \mathfrak{M}_{2q}(X). \tag{73}$$

Then

$$\|R+E\| \le \mathfrak{M}_{2q}(X)^{1/(2q)}, \tag{74}$$
$$\kappa_M \le 1+\mathfrak{M}_{2q}(X)^{1/(2q)}, \tag{75}$$
$$\alpha_M \ge \left(1-\mathfrak{M}_{2q}(X)^{1/(2q)}\right)_+. \tag{76}$$

If

$$\mathfrak{M}_{2q}(X) \le X^{2q\nu_H+o(q)},$$

then

$$\|R+E\| \le X^{\nu_H+o(1)}.$$

Proof. The defect is Hermitian, so

$$\|R+E\|^{2q} \le \operatorname{tr}(R+E)^{2q}.$$

The native Gram is $I+R+E$; its extremal eigenvalues give the next two inequalities. □

Premise (73) concerns the *complete* moment. Bounds for $\mathfrak{L}_{2q}$ or $S_{2q,2}$ do not by themselves supply it. In particular, the positivity of $S_{2q,2}$ is a structural restriction, not an upper estimate of the correct size.

10.4 Mass, degree, and represented comparison

Let $\mathcal{D}_{\mathrm{ref}}$ be a nonnegative quadratic form. Suppose, in one explicitly stated uniform or distinguished scope, that the following three estimates hold for every coefficient vector ζ in that same scope:

$$\kappa_M \le X^{\nu_M+o(1)}, \tag{77}$$
$$\langle D_M\zeta,\zeta\rangle \le X^{-\sigma_M+o(1)}\mathcal{D}_{\mathrm{ref}}(\zeta), \tag{78}$$
$$\|A_X\zeta\|^2 \ge X^{-\lambda_{\mathrm{rep}}+o(1)}\mathcal{D}_{\mathrm{ref}}(\zeta). \tag{79}$$

Then the optimal missing-native upper estimate gives the universally valid comparison

$$\|M_X\zeta\|^2 \le X^{\nu_M+\lambda_{\mathrm{rep}}-\sigma_M+o(1)}\|A_X\zeta\|^2. \tag{80}$$

Thus the familiar zero-cost condition remains

$$\boxed{\sigma_M > \nu_M+\lambda_{\mathrm{rep}}.} \tag{81}$$

The present paper changes only how ν_M might be bounded. It does not prove (77), (78), or (79).

10.5 The shorted reassembly penalty remains separate

Let A be the represented map, M the missing map, and $B=A+M$. Writing $N_A=\operatorname{Ker} A$, the represented comparison must remove

$$W_{\mathrm{nuis}} = B(N_A) = M(N_A)$$

from the full output. TPC–68 expresses the exact shorted Gram as

$$G_{\mathrm{sh}} = G_{\mathrm{dir}} - C_N, \qquad C_N \succeq 0. \tag{82}$$

The defect $R+E$ studied here is the atomic-normalized *missing-native* Gram defect. It is not automatically the direct represented-normalized defect in G_{dir}. A separate physical mass comparison is required to transfer a bound, and a separate estimate

$$\|C_N\| = o(1)$$

is required to make the nuisance shorting penalty negligible.

Consequently, even the hypothetical estimate

$$\|R+E\| = o(1)$$

would not by itself prove the endpoint reassembly theorem. It would close one native gate and leave the missing-mass, represented-frame, and nuisance gates.

10.6 Endpoint accounting

All identities proved here cost zero fixed power of X. An eventual arithmetic estimate may introduce:

$$\lambda_{\text{native}}, \qquad \lambda_{\text{mass}}, \qquad \lambda_{\text{rep}}, \qquad \lambda_{\text{nuis}},$$

or equivalent losses in the upstream notation. The complete literal loss, after signs and quantifier scopes have been matched, must remain strictly below

$$\frac{1}{400}.$$

Equality is not enough. A result outside that budget can still be a valid operator or arithmetic theorem, but it cannot be promoted to the current endpoint implication.

11 Claims, stops, and the next arithmetic door

11.1 L0 results proved here

For finite atomic-normalized native incidences:

(1) the same-residual and erasure defects decompose exactly over complete native keys:

$$R = \sum_i R_i, \qquad E = \sum_i E_i;$$

(2) R_i is residual-block diagonal and E_i is off block, giving

$$\operatorname{tr}(E_i P(R_i)) = 0;$$

(3) the first mixed trace is the cross-key rectangle sum

$$\operatorname{tr}(ER) = \sum_{i \neq j} \operatorname{tr}(E_i R_j);$$

(4) the rectangle coefficient is invariant under independent row and native-key unit gauges;

(5) every erasure-count sector $S_{k,b}$ has the exact cyclic composition formula (32);

(6) the complete two-erasure sector in every even moment is nonnegative, with the divided-difference formula (54).

11.2 L1 identifications

The matrices above are identified with the actual TPC fixed-shift interface by retaining:

- the one fixed nonzero shift h_0;
- every target $mj + h_0$, largest prime p, cofactor r, and residual label $s = d_m r$;
- every complete native key, including side, external block, opposite row, γ, and physical orbit time;
- every literal complex coefficient β_w;
- the raw row mass $W_M(m)$; and
- every actual missing and first-failure condition.

On that physical squarefree carrier, the exact source-row gauge gives two additional L1 identities: source-row signs telescope on every decorated closed walk, leaving the odd residual boundary (42), and a one-erasure word retains exactly one unequal-residual sign pair. The rectangle reduction (48) is their shortest instance. The direct-cell Γ-formula is L1 only under its explicitly declared identical-support specialization. The general atom formula is the unconditional L1 object.

11.3 L2 statements not proved

This paper does not prove:

- cancellation in one literal cross-key rectangle sum;
- a uniform local compensation estimate in every active row;
- a saving for $2q\,\mathrm{tr}(ER^{2q-1})$ at fixed or growing q;
- an upper bound of the required scale for the nonnegative sector $S_{2q,2}$;
- a complete even-moment bound uniform in all physical blocks;
- $\|R+E\| = X^{o(1)}$, much less $\|R+E\| = o(1)$;
- the missing-mass and represented-frame estimates in (78)–(79);
- a negligible shorting penalty C_N; or
- any parity-breaking estimate, prime-pair theorem, or twin-prime conclusion.

11.4 Kill criteria

(K1) **Incomplete key.** A calculation that removes side, external block, opposite row, γ, time, or another still-orthogonal output label estimates a different operator.

(K2) **False one-key door.** Searching for $\mathrm{tr}(E_iR_i)$ cancellation cannot advance the mixed second moment: that trace is identically zero. The first target is cross-key.

(K3) **Wrong parity statistic.** The number of erasure edges does not determine the sign. Only the odd residual boundary does.

(K4) **Quadratic negative-correction stop.** A proposal that uses the complete $S_{2q,2}$ as a negative correction is impossible: $S_{2q,2} \geq 0$. This does not kill the linear or higher sectors.

(K5) **Global-to-local stop.** A small $\mathrm{tr}(ER)$, an averaged row estimate, or a normalized trace does not prove the uniform local star bound.

(K6) **Arithmetic scope mismatch.** A fixed-pair theorem, an average over keys or shifts, a density-one statement, or a random sign model does not give the required complete growing-family estimate at the prescribed fixed h_0.

(K7) **Moment nonuniformity.** If $q = q_X$ grows, estimates whose constants depend uncontrolledly on q do not imply (74).

(K8) **Reassembly mismatch.** The missing-native defect $R + E$ is not H_{dir}, and neither controls C_N without additional physical comparison.

(K9) **Endpoint stop.** If the combined literal loss is at least $1/400$, the present endpoint route must stop even if every intermediate theorem is correct.

11.5 The next specified door

The shortest unresolved arithmetic object is now

$$\mathrm{tr}(ER)$$

in the exact rectangle form (49). A useful next step should:

(i) census the actual two-key rectangles without compressing their native labels;

(ii) split them by the two determinant equations and residual equality/inequality pattern;

(iii) determine whether the rectangle holonomy admits a literal pairing, dispersion identity, or local star bound; and

(iv) either prove a uniform saving or construct an actual-carrier obstruction closing this linear door.

If the linear sector is structurally too large or its residual boundary freezes on the actual support, the next alternatives are higher one-erasure polygons, literal complex-phase cancellation, or direct shorted reassembly. The nonnegative quadratic sector is not an alternative source of negative compensation.

11.6 Status line

TPC–69 moves the signed operator route from edgewise Möbius products to the first exact cross-key holonomy and proves a new positivity obstruction for the quadratic erasure response. This is a genuine L0/L1 refinement of the proof architecture. It is not an L2 cancellation theorem and provides no evidence that the parity barrier has been broken.

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